

MAE 3128, Spring 2026

Homework assignment 2

Professor Kartik Bulusu

February 13, 2026

This assignment is due at the end of the day on Wednesday, February 25, 2026. Collaboration is encouraged.

Solutions will be graded on the following criteria: Correct answer, correct units, appropriate significant digits, problem statement and assumptions, and neatness.

HW2 P1: What is the maximum shear stress, in terms of the applied load F and cross-sectional area A , in a uniaxially stressed structure, and what is the associated angle α_s relative to an axial direction?

$$\tau_{xy}|_{\max} = \sqrt{\left(\frac{\sigma_{xx} - \cancel{\sigma_{yy}^0}}{2}\right)^2 + \cancel{\tau_{xy}^0}^2}$$

$$= \sqrt{\left(\frac{\sigma_{xx}}{2}\right)^2} = \frac{\sigma_{xx}}{2}$$

$$\Rightarrow \tau_{xy}|_{\max} = \frac{\sigma_{xx}}{2}$$

$$\Rightarrow \tau_{xy}|_{\max} = \frac{F}{2A}$$

$$\alpha_s = \frac{1}{2} \tan^{-1} \left(\frac{\cancel{\sigma_{yy}^0} - \sigma_{xx}}{2\cancel{\tau_{xy}^0}} \right) = \frac{1}{2} \tan^{-1}(\infty)$$

$$\alpha_s = 45^\circ \text{ or } \frac{\pi}{4} \text{ rad}$$

HW2 P2: Referring to Example 3.3 (also Course Work 4 problem 2), compute the maximum shear stress $\sigma'_{z\theta}$ of $f = 0$; that is,

$$\sigma_{\theta\theta} = \frac{Pa}{h}, \sigma_{zz} = \frac{Pa}{2h}$$

Compare it to the case when

$$f = Pa^2\pi.$$

When $f = 0$:

$$\sigma_{z\theta} \Big|_{\max, \min} = \pm \sqrt{\left(\frac{\sigma_{\theta\theta} - \sigma_{zz}}{2}\right)^2 + \cancel{\sigma_{z\theta}^2}} \quad \text{neglect shear stress}$$

$$\Rightarrow \sigma_{z\theta} \Big|_{\max} = \left(\frac{\sigma_{\theta\theta} - \sigma_{zz}}{2}\right) = \frac{1}{2} \left(\frac{Pa}{h} - \frac{Pa}{2h}\right)$$

$$\Rightarrow \boxed{\sigma_{z\theta} \Big|_{\max} = \frac{Pa}{4h}} \quad \text{orientation: } \alpha_s = \frac{1}{2} \tan^{-1} \left(\frac{\sigma_{zz} - \sigma_{\theta\theta}}{2\sigma_{z\theta}} \right)$$

$$\Rightarrow \alpha_s = \frac{1}{2} \tan^{-1}(\infty)$$

$$\Rightarrow \boxed{\alpha_s = 45^\circ \text{ or } \frac{\pi}{4} \text{ rad}}$$

When $f = Pa^2\pi$:

$$\sigma_{z\theta} \Big|_{\max} = \left(\frac{\sigma_{\theta\theta} - \sigma_{zz}}{2}\right) \quad \left[\begin{array}{l} \sigma_{\theta\theta} = \frac{Pa}{h} \\ \sigma_{zz} = \frac{Pa}{2h} + \frac{f}{2\pi ah} \end{array} \right]$$

$$\Rightarrow \sigma_{z\theta} \Big|_{\max} = \frac{1}{2} \left[\frac{Pa}{h} - \frac{Pa}{2h} - \frac{f}{2\pi ah} \right]$$

$$= \frac{1}{2} \left[\frac{Pa}{h} - \frac{Pa}{2h} - \frac{Pa^2\pi}{2\pi ah} \right] = \frac{1}{2} \left[\cancel{\frac{Pa}{h}} - \frac{Pa}{2h} \right]$$

$$\Rightarrow \boxed{\sigma_{z\theta} \Big|_{\max} = 0}$$

HW2 P3: Design and experiment to determine the failure strength of a chordae tendineae. Additionally, outline a method of analysis to estimate the range of stress experienced by the chordae in vivo under physiologic values of intraventricular pressure.

↳ Answers vary

↳ Points awarded for clear and concise explanation

HW2 P4: Find the dimensions of and pressures within the human interior vena cava. Estimate the mean wall stress $\sigma_{\theta\theta}$. Even though the aorta is not thin walled, estimate its mean wall stress $\sigma_{\theta\theta}$ in the human abdominal aorta. Discuss structural differences between the vena cava and aorta given their different function and mechanical environments.

From Appendix A9.2			
	Inner radius (cm)	Wall thickness (mm)	Pressure (mm Hg)
Vena cava	1.5 cm	1.5	10
Aorta	1.25	2	120/80

↳ Answers vary

↳ Points given for analysis related to $\sigma_{\theta\theta}$ stress or equivalent with explanation

HW2 P5: Although Eq. 3.41 ($\sigma_{\theta\theta} = \frac{Pa}{h}$) was derived for a thin-walled cylinder, it can be shown that it provides a reasonable estimate of the mean circumferential stress in a thick-walled tube as well. Therefore, in hypertension (high blood pressure), the aorta distends (a increases) and the wall thins (h decreases). Thus, $\sigma_{\theta\theta}$ increases tremendously. If the hypertensive pressure is $P_H = nP$, where n is a number, and the luminal radius returns to a , how much does the aorta need to thicken to restore $\sigma_{\theta\theta}$ to its original value?

$$\sigma_{\theta\theta} = \frac{Pa}{h}$$

$$P_H = P_{\text{hypertension}} = nP$$

$$h_H = h_{\text{hypertension}}$$

$$\Rightarrow \frac{Pa}{h} = \frac{nPa}{h_H} \Rightarrow \boxed{h_H = nh}$$

HW2 P6: In the thick-walled cylinder problem, we found that $\sigma_{zz} = \nu(\sigma_{rr} + \sigma_{\theta\theta})$. If we use this result in Hooke's law, then

$$\epsilon_{rr} = \frac{1}{E} \left\{ \sigma_{rr} - \nu [\sigma_{\theta\theta} + \nu(\sigma_{rr} + \sigma_{\theta\theta})] \right\}$$

$$\epsilon_{\theta\theta} = \frac{1}{E} \left\{ \sigma_{\theta\theta} - \nu [\sigma_{rr} + \nu(\sigma_{rr} + \sigma_{\theta\theta})] \right\}$$

Show that these two equations can be inverted to yield

$$\sigma_{rr} = \frac{E}{(1 + \nu)(1 - 2\nu)} \left[(1 - \nu)\epsilon_{rr} + \nu\epsilon_{\theta\theta} \right]$$

$$\sigma_{\theta\theta} = \frac{E}{(1 + \nu)(1 - 2\nu)} \left[\nu\epsilon_{rr} + (1 - \nu)\epsilon_{\theta\theta} \right]$$

HW2 P6

Given, $\sigma_{zz} = \nu(\sigma_{rr} + \sigma_{\theta\theta})$

$$\begin{aligned} \textcircled{1} \quad \epsilon_{rr} &= \frac{1}{E} \{ \sigma_{rr} - \nu[\sigma_{\theta\theta} + \nu(\sigma_{rr} + \sigma_{\theta\theta})] \} \\ \textcircled{2} \quad \epsilon_{\theta\theta} &= \frac{1}{E} \{ \sigma_{\theta\theta} - \nu[\sigma_{rr} + \nu(\sigma_{rr} + \sigma_{\theta\theta})] \} \end{aligned}$$

Show :

$$\sigma_{rr} = \frac{E}{(1+\nu)(1-2\nu)} [(1-\nu)\epsilon_{rr} + \nu\epsilon_{\theta\theta}]$$

$$\sigma_{\theta\theta} = \frac{E}{(1+\nu)(1-2\nu)} [\nu\epsilon_{rr} + (1-\nu)\epsilon_{\theta\theta}]$$

$$\begin{aligned} E\epsilon_{rr} &= \sigma_{rr} - \nu\sigma_{\theta\theta} - \nu^2\sigma_{rr} - \nu^2\sigma_{\theta\theta} \quad \leftarrow \text{From (1)} \\ &= \sigma_{rr}(1-\nu^2) - \sigma_{\theta\theta}(\nu+\nu^2) \rightarrow (3) \end{aligned}$$

$$\begin{aligned} E\epsilon_{\theta\theta} &= \sigma_{\theta\theta} - \nu\sigma_{rr} - \nu^2\sigma_{rr} - \nu^2\sigma_{\theta\theta} \quad \leftarrow \text{From (2)} \\ &= \sigma_{\theta\theta}(1-\nu^2) - \sigma_{rr}(\nu+\nu^2) \rightarrow (4) \end{aligned}$$

Let $A = (1-\nu^2)$ & $B = (\nu+\nu^2) = \nu(1+\nu)$

$$\begin{aligned} \text{From (3)} \quad E\epsilon_{rr} &= A\sigma_{rr} - B\sigma_{\theta\theta} \\ \text{From (4)} \quad E\epsilon_{\theta\theta} &= A\sigma_{\theta\theta} - B\sigma_{rr} \end{aligned} \quad \left\{ \begin{array}{l} \text{Multiplying} \\ \text{with A} \\ \text{with B} \end{array} \right.$$

$$\textcircled{5} \quad E(A\epsilon_{rr} + B\epsilon_{\theta\theta}) = \sigma_{rr}(A^2 - B^2)$$

From (5) $\sigma_{rr} = \frac{E(A\epsilon_{rr} + B\epsilon_{\theta\theta})}{A^2 - B^2} = \frac{E(A\epsilon_{rr} + B\epsilon_{\theta\theta})}{(A+B)(A-B)}$

$$\Rightarrow \sigma_{rr} = \frac{E[(1-\nu^2)\epsilon_{rr} + (\nu+\nu^2)\epsilon_{\theta\theta}]}{(1-\nu^2 + \nu + \nu^2)(1-\nu^2 - \nu - \nu^2)} \quad \left\{ \begin{array}{l} \because \\ a^2 - b^2 = (a+b)(a-b) \end{array} \right.$$

$$= \frac{E[(1-\nu)(1+\nu)\epsilon_{rr} + \nu(1+\nu)\epsilon_{\theta\theta}]}{(1+\nu)(1-\nu-2\nu^2)}$$

$$= \frac{E(1+\nu)[(1-\nu)\epsilon_{rr} + \nu\epsilon_{\theta\theta}]}{(1+\nu)(1-\nu-2\nu^2)}$$

$$\sigma_{rr} = \frac{E[(1-\nu)\epsilon_{rr} + \nu\epsilon_{\theta\theta}]}{(1+\nu)(1-2\nu)} \rightarrow \textcircled{6}$$

$$\left\{ \begin{array}{l} (1+\nu)(1-2\nu) \\ 1+\nu-2\nu-2\nu^2 \\ 1-\nu-2\nu^2 \end{array} \right.$$

$$\text{Let } A = (1 - \nu^2) \quad B = (\nu + \nu^2) = \nu(1 + \nu)$$

$$\begin{aligned} \text{From (3)} \quad \left. \begin{aligned} E \epsilon_{rr} &= A \sigma_{rr} - B \sigma_{\theta\theta} \\ E \epsilon_{\theta\theta} &= A \sigma_{\theta\theta} - B \sigma_{rr} \end{aligned} \right\} \Rightarrow \begin{aligned} B E \epsilon_{rr} &= A B \sigma_{rr} - B^2 \sigma_{\theta\theta} \quad \leftarrow \text{Multiplying with } B \\ A E \epsilon_{\theta\theta} &= A^2 \sigma_{\theta\theta} - A B \sigma_{rr} \quad \leftarrow \text{Multiplying with } A \end{aligned} \end{aligned}$$

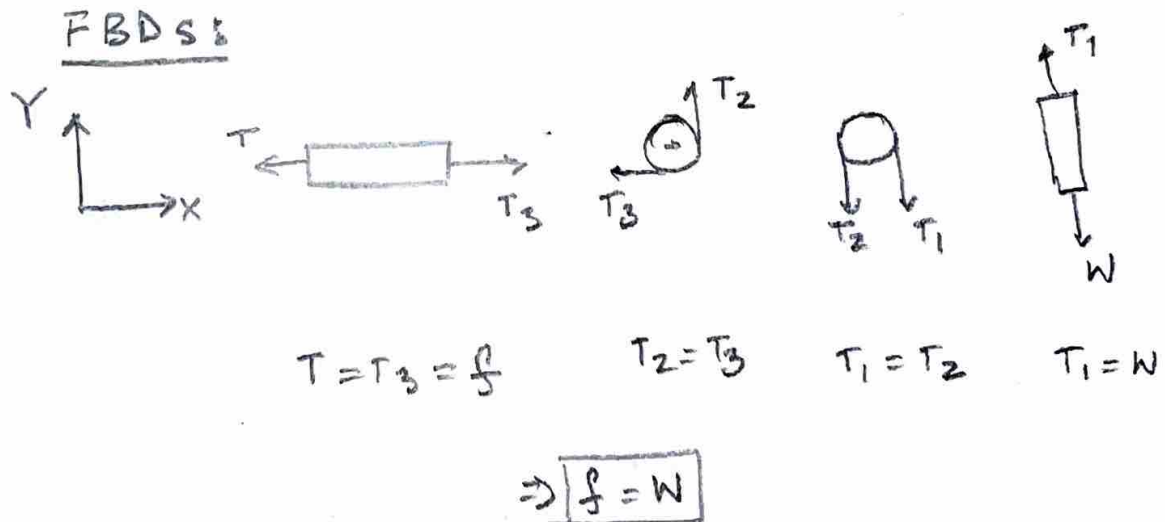
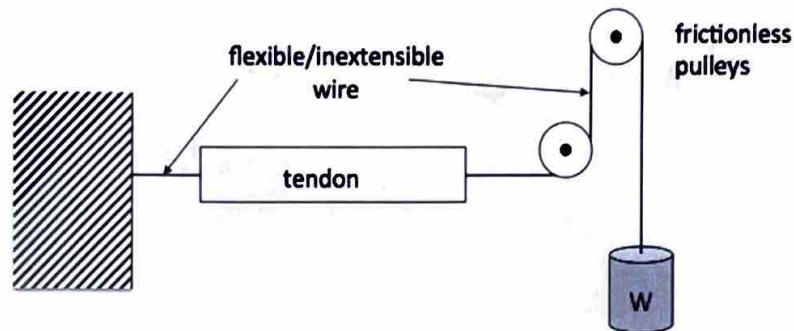
$$E(A \epsilon_{\theta\theta} + B \epsilon_{rr}) = \sigma_{\theta\theta} (A^2 - B^2) \Rightarrow (7)$$

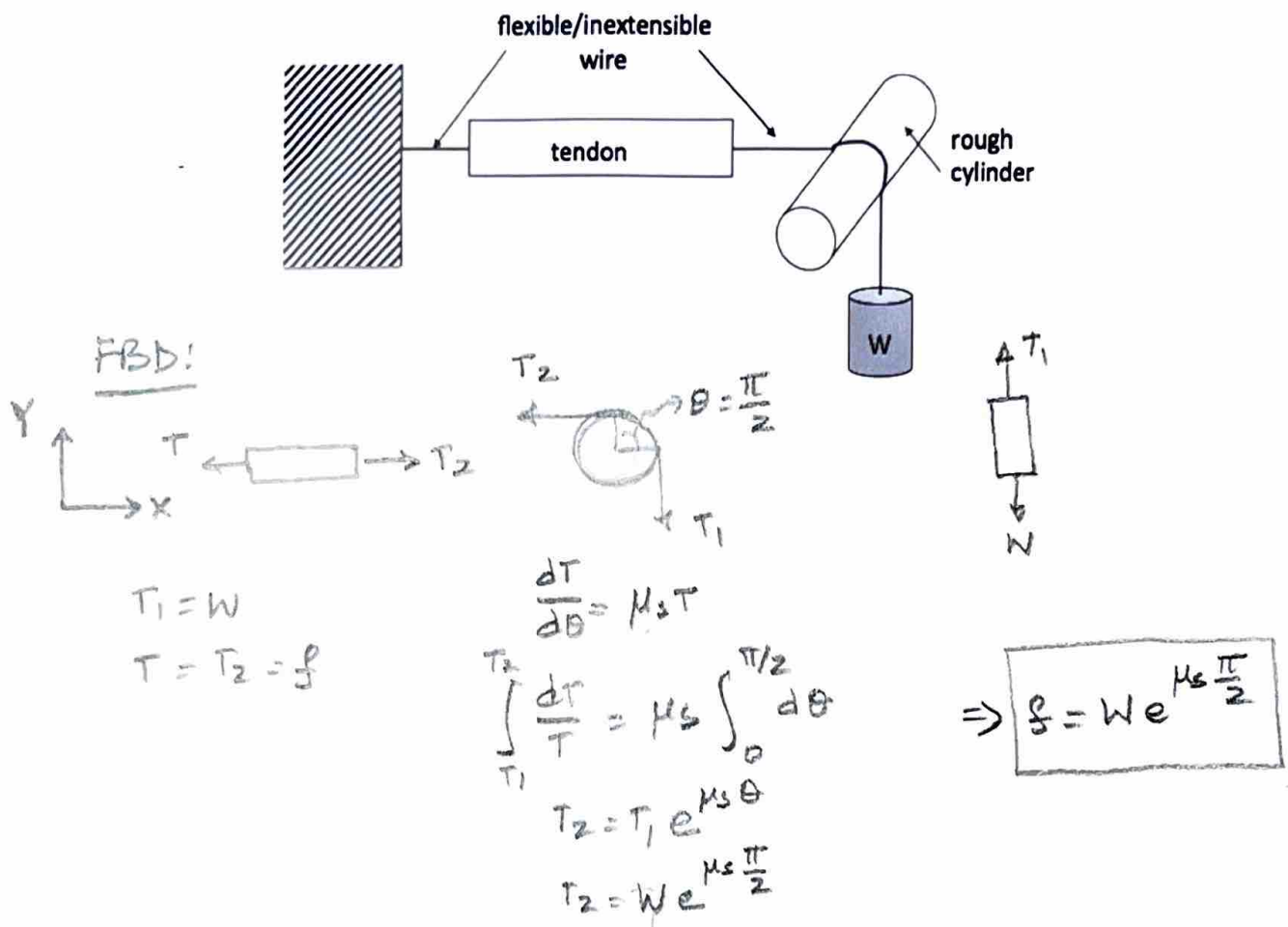
$$\text{From (7)} \quad \sigma_{\theta\theta} = \frac{E(A \epsilon_{\theta\theta} + B \epsilon_{rr})}{A^2 - B^2} = \frac{E(A \epsilon_{\theta\theta} + B \epsilon_{rr})}{(A+B)(A-B)}$$

$$\begin{aligned} \Rightarrow \sigma_{\theta\theta} &= \frac{E[(1-\nu^2)\epsilon_{\theta\theta} + (\nu+\nu^2)\epsilon_{rr}]}{(1-\nu^2 + \nu+\nu^2)(1-\nu^2 - \nu-\nu^2)} \\ &= \frac{E[(1-\nu)(1+\nu)\epsilon_{\theta\theta} + \nu(1+\nu)\epsilon_{rr}]}{(1+\nu)(1-\nu-2\nu^2)} \\ &= \frac{E(1+\nu)[(1-\nu)\epsilon_{\theta\theta} + \nu\epsilon_{rr}]}{(1+\nu)(1-\nu-2\nu^2)} \end{aligned}$$

$$\boxed{\sigma_{\theta\theta} = \frac{E[\nu \epsilon_{rr} + (1-\nu)\epsilon_{\theta\theta}]}{(1+\nu)(1-2\nu)}} \quad \checkmark$$

HW2 P7: Consider the two experimental setups shown below. They are both designed to impose an axial load on a thin tendon for the purposes of studying the stress-strain behavior. Assuming a coefficient of friction μ_s between the wire and the rough cylinder, determine the axial load f that is applied to the tendon, in terms of W for each case.





HW2 P8: If the nominal axial stress in the tendon experiments from Problem 7 is $\sigma_o = f/A_o$, what is the axial Cauchy stress σ if the tendon has an original cross-sectional area A_o and length L , and the tendon is incompressible? Let the current area and length be A and l , respectively.

Incompressible materials conserve volume:

$$A_o L = A l \Rightarrow A = A_o \frac{L}{l}$$

$$\sigma = \frac{f}{A} = \frac{f}{A_o L/l} = \frac{l}{L} \frac{f}{A_o}$$

$$\sigma = \frac{l}{L} \sigma_o$$